

PROJECT REPORT

MySQL Employee Database-Querying and Analysis

Retrieving, Filtering, Sorting, Grouping, and Analyzing Employee Data

1. Project Overview

This project demonstrates the use of SQL queries to retrieve, filter, sort, group, and analyze employee data in MySQL. It was completed as part of a hands-on assignment covering core querying techniques and multi-table joins, using an employee database consisting of employees, departments, and location tables.

The following SQL operations were performed over the course of this assignment:

- DISTINCT
- Aliases using AS
- WHERE clause and operators
- UPDATE
- ORDER BY
- LIMIT
- Aggregate Functions
- GROUP BY
- HAVING
- INNER JOIN
- LEFT JOIN
- RIGHT JOIN

2. SQL Queries Performed

2.1 DISTINCT Values

Retrieve distinct salaries from the Employees table

```
SELECT DISTINCT salary
FROM employees;
```

2.2 Alias (AS)

Provide aliases for age and salary

```
SELECT age AS Employee_Age,
       salary AS Employee_Salary
FROM employees;
```

2.3 WHERE Clause & Operators

Retrieve employees with a salary greater than ₹50,000 and hired before January 1, 2016

```
SELECT employee_name, salary, hire_date
FROM employees
WHERE salary > 50000
      AND hire_date < '2016-01-01';
```

Find the employee whose designation is missing

```
SELECT *
FROM employees
WHERE designation IS NULL;
```

Update the missing designation to "Data Scientist"

```
UPDATE employees
SET designation = 'Data Scientist'
WHERE designation IS NULL;
```

2.4 Sorting and Grouping Data

ORDER BY — Sort employees by Department ID in ascending order and Salary in descending order

```
SELECT employee_name, department_id, salary
FROM employees
ORDER BY department_id ASC, salary DESC;
```

LIMIT — Display the first 5 employees hired in the year 2018

```
SELECT employee_name, hire_date
FROM employees
WHERE YEAR(hire_date) = 2018
LIMIT 5;
```

2.5 Aggregate Functions

Calculate the sum of all salaries in the Finance department

```
SELECT SUM(salary) AS Total_Salary
FROM employees
WHERE department_id = 7;
```

Note: Update the department_id based on the ID assigned to the Finance department in your table.

Find the minimum age among all employees

```
SELECT MIN(age) AS Minimum_Age
FROM employees;
```

2.6 GROUP BY

List the maximum salary for each location

```
SELECT location_id,
       MAX(salary) AS Maximum_Salary
FROM employees
GROUP BY location_id;
```

Calculate the average salary for each designation containing the word "Analyst"

```
SELECT designation,
       AVG(salary) AS Average_Salary
FROM employees
WHERE designation LIKE '%Analyst%'
GROUP BY designation;
```

2.7 HAVING

Find departments with less than 3 employees

```
SELECT department_id,
       COUNT(*) AS Employee_Count
FROM employees
GROUP BY department_id
HAVING COUNT(*) < 3;
```

Find locations with female employees whose average age is below 30

```
SELECT location_id,
       AVG(age) AS Average_Age
FROM employees
WHERE gender = 'Female'
GROUP BY location_id
HAVING AVG(age) < 30;
```

3. Joins

3.1 INNER JOIN

List employee names, their designations, and department names where employees are assigned to a department

```
SELECT e.employee_name,  
       e.designation,  
       d.department_name  
FROM employees e  
INNER JOIN departments d  
ON e.department_id = d.department_id;
```

3.2 LEFT JOIN

List all departments along with the total number of employees in each department

This query also includes departments with no employees.

```
SELECT d.department_id,  
       d.department_name,  
       COUNT(e.employee_id) AS Employee_Count  
FROM departments d  
LEFT JOIN employees e  
ON d.department_id = e.department_id  
GROUP BY d.department_id, d.department_name;
```

3.3 RIGHT JOIN

Display all locations along with employee names assigned to each location

Locations without employees will display NULL for the employee name.

```
SELECT l.location_id,  
       l.location,  
       e.employee_name  
FROM employees e  
RIGHT JOIN location l  
ON e.location_id = l.location_id;
```

4. Key SQL Concepts Practiced

The table below summarizes the core SQL concepts applied throughout this assignment and the purpose each one serves.

SQL Concept	Purpose
DISTINCT	Removes duplicate values
AS	Provides aliases for columns
WHERE	Filters rows based on conditions
UPDATE	Modifies existing data
ORDER BY	Sorts query results
LIMIT	Restricts the number of rows returned
SUM()	Calculates the total
MIN()	Finds the minimum value
MAX()	Finds the maximum value
AVG()	Calculates the average
COUNT()	Counts records
GROUP BY	Groups rows with similar values
HAVING	Filters grouped results
INNER JOIN	Returns matching records from both tables
LEFT JOIN	Returns all records from the left table
RIGHT JOIN	Returns all records from the right table

5. Learning Outcome

Through this assignment, I practiced querying and analyzing data using MySQL. I gained hands-on experience with:

- Filtering and sorting data
- Aggregate functions
- Grouping data using GROUP BY
- Filtering grouped data using HAVING
- Using WHERE conditions
- Working with INNER JOIN, LEFT JOIN, and RIGHT JOIN

This assignment helped strengthen my understanding of how relational databases can be queried to extract meaningful insights from employee, department, and location data.

6. Author

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